

Question 1



Figure 1-1: Google LLC's autonomous car

Since its launching in USA in 2007, Google's Street View cars (see **Figure 1-1**) have been circling the globe, using roof-mounted cameras to map more than 10 million miles of imagery. As the team lead of Google LLC's autonomous car project, you are working very hard to convert all its manual driving cars to automated driving cars, which are referred to as the agent-X. Suppose multiple versions of agent-X are implemented for experimental purposes based on five different agent design programs, which are *table-driven agent*, *simple reflex agent*, *model-based reflex agent*, *goal-based agent*, *utility-based agent*, answer the following questions:

- a) Describe in your own words what it means by *agent*. (2 marks)

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

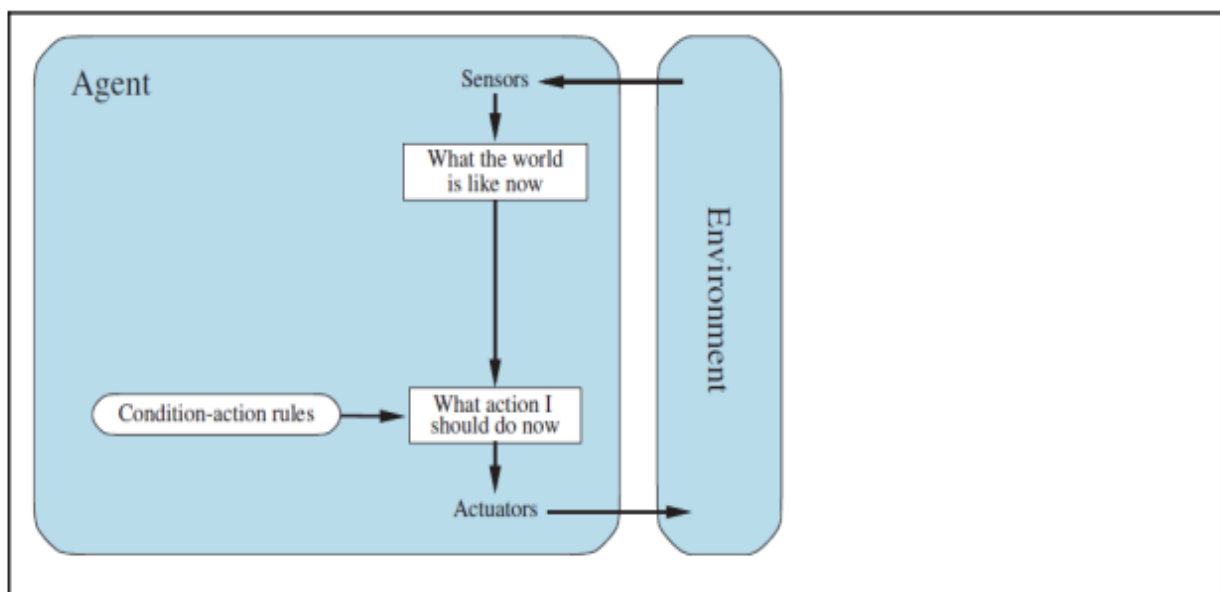
- b) A *rational agent* is a type of intelligent agent that acts based on the concept of *rationality*. Briefly describe the **FOUR (4)** components of *rationality*. Then, describe in your own words what it means by *rational agent*. (4 + 2 marks)

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.

- The actions that the agent can perform.
- The agent's percept sequence to date.
- A rational agent should select an action that is expected to maximize its performance measure , based on the evidence provided by the percept sequence and its built-in knowledge .

c) Explain how a *simple reflex agent* works with the aid of an *illustration*.

(4 marks)



d) Identify **THREE (3)** challenges of building a *utility-based agent*.

(3 marks)

- Agents need to model and keep track of complicated environments.
- Developing such models involves a great deal of research on perception, representation, reasoning, and learning.
- Developing utility-function requires the use of effective and efficient algorithms.
- Solutions with high computational complexity are not practical.

- e) Identify an agent program between *table-driven agent*, and *simple reflex agent* that is more suitable to be applied as the solution. Explain why your selected choice is more suitable and why another agent program is less suitable to support your answer. (1 + 4 marks)

- Simple reflex agents.
- The implementation of a table-driven agent is impractical as it requires the construction of a table of infinite size to retain the complete percept sequence in the memory .
- The implementation of a simple reflex agent is practical as it selects actions based on the current percept only, ignoring the rest of the percept history , and hence the size of its agent program is a lot smaller

- f) Identify an agent program between *goal-based agent*, and *utility-based agent* that is more suitable to be applied as the solution. Explain why your selected choice is more suitable and why another agent program is less suitable to support your answer. (1 + 4 marks)

- Utility-based agents.
- Goal-based agents are non-optimal (or are unable to deliver the best performance) , because when more than one solution is available, they can differentiate one from another .
- Utility-based agents can deliver the best expected performance , because they can tell which solution is more promising among all solutions available .

Question 2

- a) The *cutting stock problems*, as illustrated in **Figure 2-1**, occur when materials such as steel rods are processed in different lengths and diameters, among other things. It is a problem that aims to minimise material wasted, which arises in a variety of manufacturing industries such as rod industry, paper industry, wood industry, plastic, and leather manufacturing industry etc.

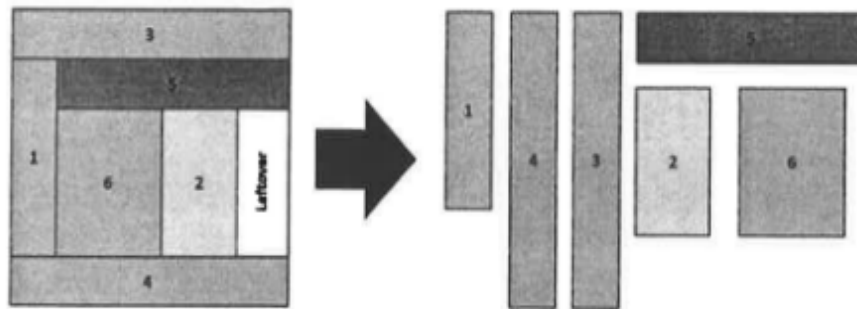


Figure 2-1: The cutting stock problems.

- (i) Determine whether the *cutting-stock problem* is an optimisation problem and justify your answer. (1 + 2 marks)

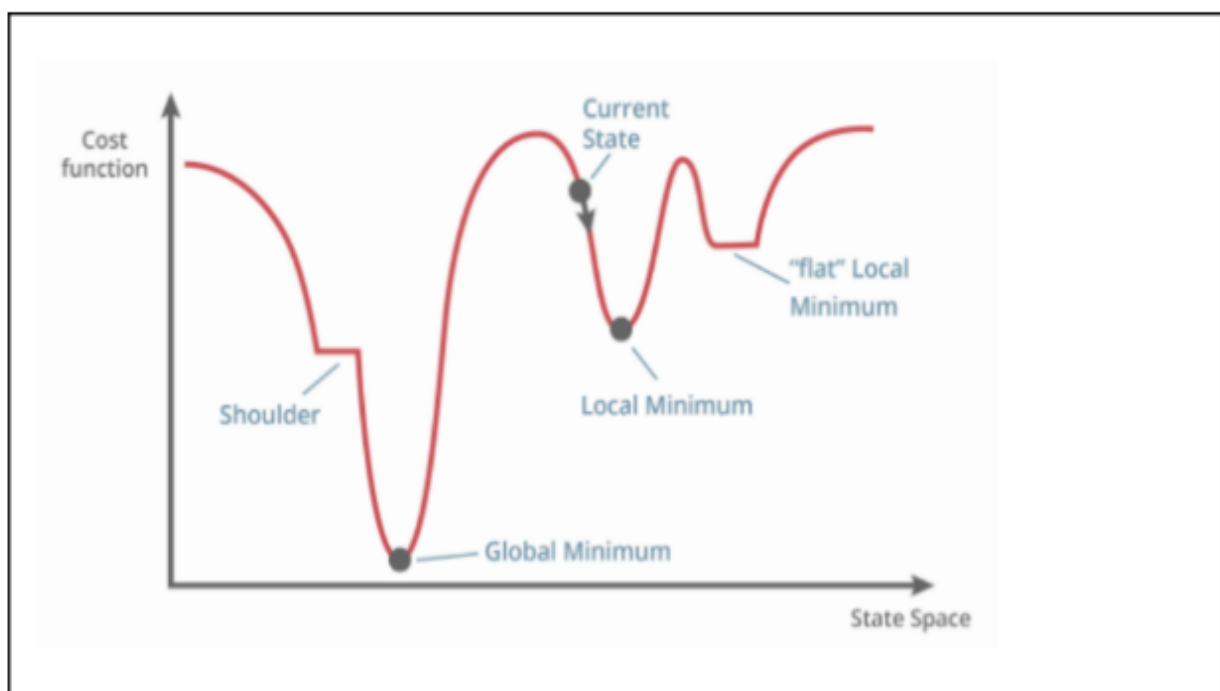
- Yes.
- The reason is that it is a problem with many sub-optimal solutions which are least preferable but it is extremely difficult to find the optimal solution.

- (ii) *Local search strategies* are usually used in solving optimisation problems. Identify **TWO** (2) differences between *local search strategies* and *classical search strategies*. (4 marks)

	Classical Search	Local Search
Solution	Return a sequence of actions as a solution.	Return a single final state as a solution.
Optimality	CSS finds an optimal solution.	LSS finds a good enough solution without guaranteeing optimality.

Space Consumption	CSS consume more space as they track both expanded and unexpanded nodes.	LSS uses less space by focusing only on the current state.
Search Pattern	CSS explores the search space systematically (like checking every possible road in order)	LSS moves heuristically between states without exhaustive exploration.

- (iii) A *state-space landscape* is a graphical representation of different states of a problem reachable by a local search. Draw a *state-space landscape* that comprises **FIVE (5)** states usually encountered by a local search in which its elevation corresponds to a *heuristic cost function*. (4 marks)



- b) The *4-queens problem* is a problem of placing four chess queens on a 4×4 chessboard so that no two queens threaten each other; thus, a solution requires that no two queens share the same row, column, or diagonal. Suppose a one-column one-queen formulation is used to solve a problem as illustrated in **Figure 2-2**, where the heuristic cost function h is defined as *the number of pairs of queens that are attacking each other, either directly or indirectly*, answer the following three questions based on information provided in **Figure 2-2**:



Figure 2-2: An example of a 4×4 chessboard.

- (i) Calculate the number of successors for the problem. (1 mark)

Total number of successors per state = $4 \times 3 = 12$

- (ii) Calculate the h -value of the state of the problem provided in **Figure 2-2**. (1 mark)

h -value = 1

- (iii) Redraw the chessboard provided in **Figure 2-2** by filling up all the unoccupied squares with the proper h -values. (12 marks)

		4	3
3	2	3	
0	1	3	3
1	2		3

Question 3

International Conference on Digital Transformation and Applications (ICDXA) is a conference organised by Tunku Abdul Rahman University of Management and Technology (TAR UMT) in 2024 that aims to promote research, education, and industry implementation that involve Digital Transformation (DX) processes. Suppose the conference will happen in two months' time, and you are assigned the task of allocating five events to three venues based on information provided in **Table 3-1** and **Table 3-2**, and each event will only be allocated one venue at a time, answer the following questions:

Table 3-1: Event.

No.	Event Code	Track	Time
1	P01-0001	Speech by industry speakers	9:00am – 12:00pm
2	P01-0002	Full paper presentation	10:00am – 1:00pm
3	P01-0003	Abstract presentation	11:00am – 2:00pm
4	P01-0004	Final year project exhibition	12:00pm – 3:00pm
5	P01-0005	Artificial Intelligence (A.I.) workshop	2:00pm – 5:00pm

Table 3-2: Availability of venues based on capacity.

No.	Lecturer	Events
1	Alan Turing	2, 3, 4
2	Steve Jobs	1, 2, 4, 5
3	Dennis Ritchie	1, 4, 5

- a) Formulate this problem as a *constraint satisfaction problem (CSP)* in which there is one *variable* per class, by stating *domains* for each variable (after the enforcement of unary constraints) and *constraints* in a format that represents *abstract relations*. (9 marks)

Variables = {E1, E2, E3, E4, E5}

Variables Domains

E1 {SJ, DR}

E2 {AT, SJ}

E3 {AT}

E4 {AT, SJ, DR}

E5 {SJ, DR}

Constraints = {E1 ≠ E2, E1 ≠ E3, E2 ≠ E3, E2 ≠ E4, E3 ≠ E4, E4 ≠ E5}

- b) Design a *constraint graph* associated with the CSP.

(6 marks)



- c) Name the most popular algorithm applied for enforcing arc consistency.

(1 mark)

AC-3 algorithm

- d) Identify domains of the variables after running arc-consistency on this initial graph based on the assumption that all unary constraints have already been enforced.

(5 marks)

Variables	Domains
E1	{DR} (1%)
E2	{SJ} (1%)
E3	{AT} (1%)
E4	{DR} (1%)
E5	{SJ} (1%)

- e) Referring to the answer for **Question 3 d)**, identify the number of solutions available to this CSP. Then, write down **ONE** (1) solution.

(1 + 1 marks)

- 1
- The solution is $\{E1 = DR, E2 = SJ, E3 = AT, E4 = DR, E5 = SJ\}$

f) Identify whether your *CSP* solution for **Question 3 e)** is *consistent* or *inconsistent*, and whether it is *complete* or *partial*. (2 marks)

Consistent and complete